



HMI Stars: Web & Mobile Application

Design and development of a real-time communication and management ecosystem for **HMI Stars Consulting**

FINAL YEAR PROJECT · 2025/2026

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The Problem: Fragmented Operations

Before this project, HMI Stars Consulting relied on **email, instant messaging, and physical paperwork** — resulting in critical operational failures:

Lost Documents

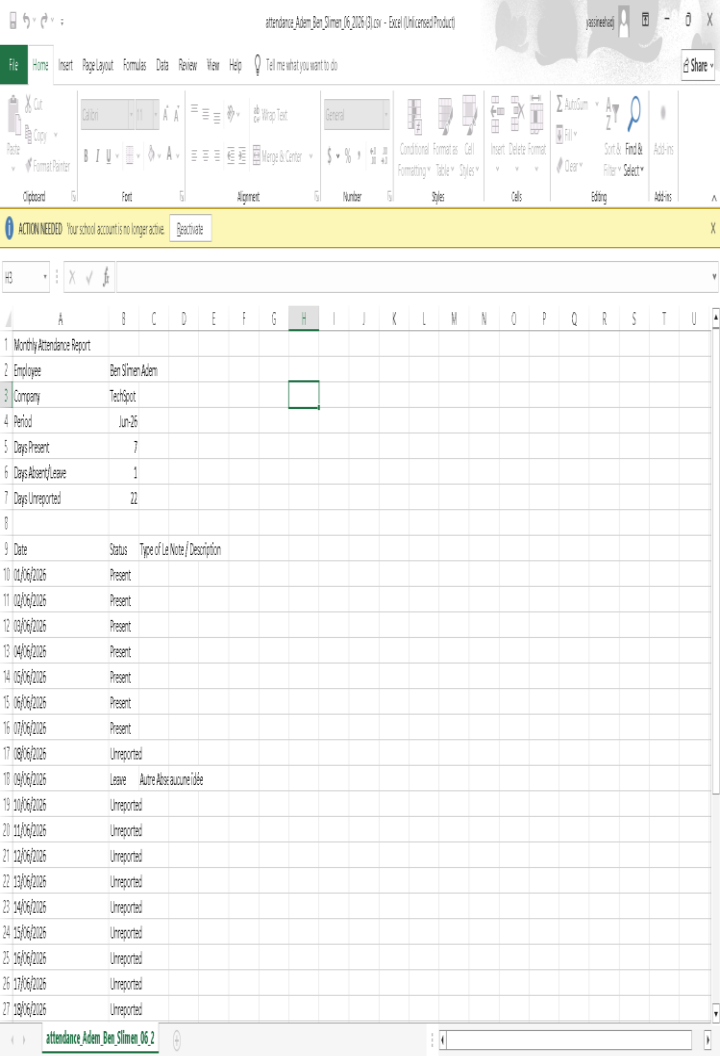
KBIS records and VAT declarations misplaced in cluttered email threads

No Traceability

Employee attendance logs lacked centralised tracking or audit trails

Delayed Response

Urgent administrative requests averaged **3–5 business days** to resolve



Challenge 1: Manual Overhead

Operational overhead caused by paper logging and spreadsheet processing makes **payroll tracking and verification** slow and error-prone:

Manual Logging

Errors, late timesheets, and lost check-ins leading to operational delays.

Slow Processing

End-of-week payroll validation bottlenecks and delayed reviews.

No History

Lack of immediate lookup logs and **digital records** for compliance audits.



HMI Stars

Choisissez votre entreprise

Votre compte est lié à 3 entreprises.
Sélectionnez celle que vous souhaitez gérer.

**TechSpot**

SIRET: 12345678912345
sas

**Test1**

SIRET: 566516151
sas

**test2**

SIRET: 6515613513
sas



 Se déconnecter

Challenge 2: Administrative Data Silos

Isolated employee logs and document tracking across separate
communication platforms and spreadsheets risk data security:

Isolated Info

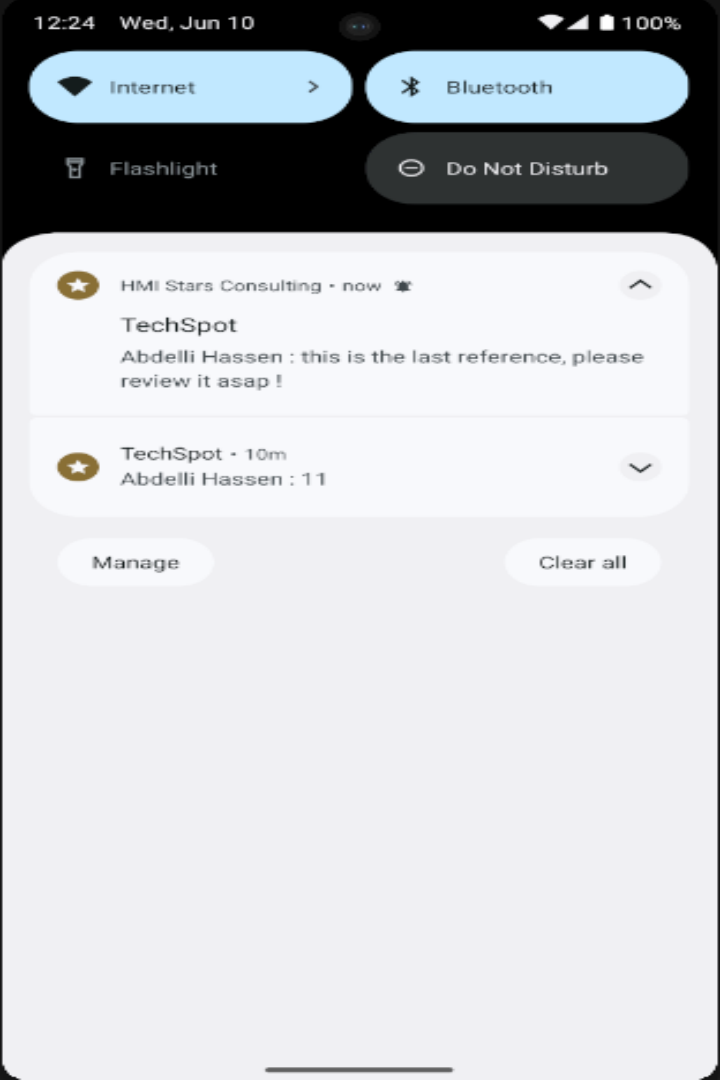
Employee profiles, attendance logs, and contracts stored separately.

Communication Silos

Documents and files shared via unstructured chats and email threads.

Sync Overhead

Manual copies required to **synchronize tables** across departments.



Challenge 3: Offline Disconnections

Remote field construction sites frequently lose cellular network **coverage**, causing **standard client systems** to fail:

Outage Areas

Basements, remote yards, and warehouses lose cellular signal.

Web Crashes

Standard web forms fail to load without active internet connection.

Data Loss

Timesheet entries and clock-in logs **get lost on** connection drops.

Project Objectives

Establishing the system's core operational and technical requirements:



Unified Hub

Centralize files and chats in one spot for immediate audit lookup.



Offline Sync

Implement local database buffers on mobile clients for network dropouts.

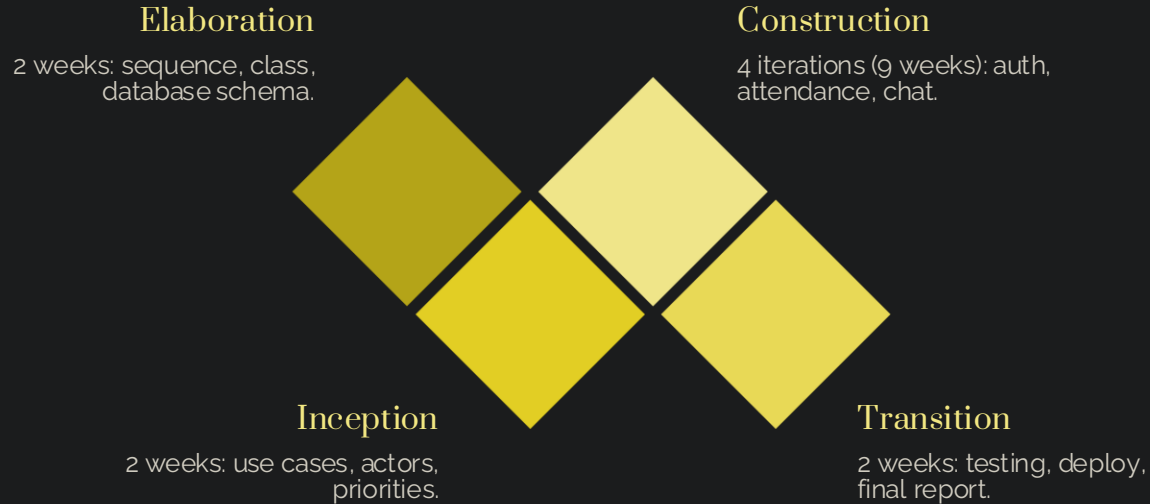


Zero Trust Security

Parse engine-level RLS policies to isolate tenant data.

Methodology: 2TUP

The **Two Track Unified Process (2TUP)** separates development into a **functional track** (requirements, use cases) and a **technical track** (architecture, database). Both converge during Construction.



Each Construction iteration targeted specific use cases, enabling incremental validation of RLS policies, state management, and offline caching before final integration.

Functional Actors

Defining platform roles and database access boundaries:



Client Manager

Registers daily check-ins, files leaves, and uploads docs using mobile app.



HMI Manager

Verifies company accounts, audits legal files, and manages users on web portal.



Automation Engine

Parses PostgreSQL security, processes file archives, and routes WS notifications.

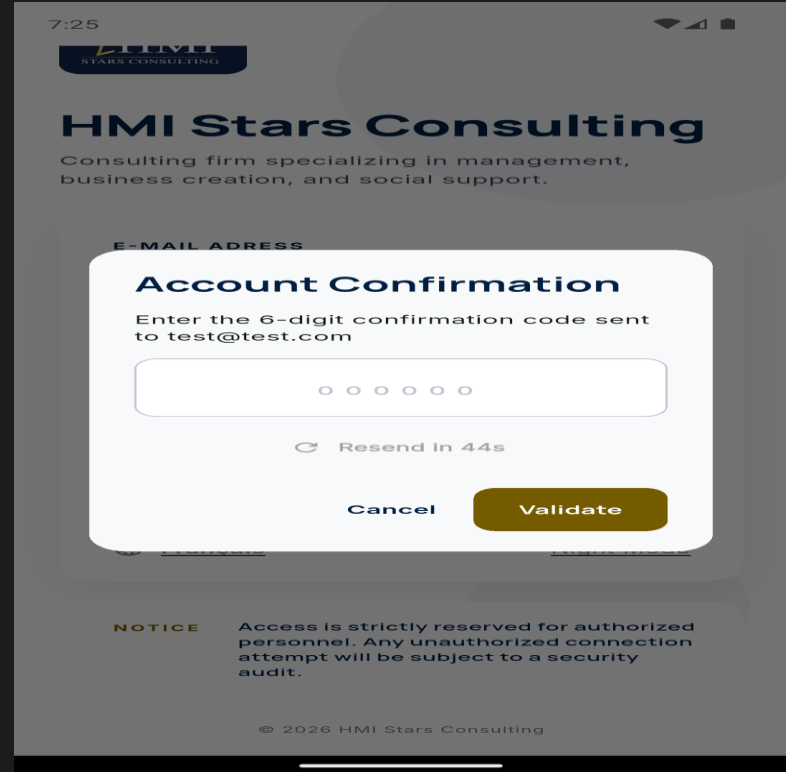
Ecosystem Use Cases

Core Operations

Authentication: Secure magic link redirect and passwordless OTP sign-in.
Core Records: Employee registry construction and timesheet tracking.

Support Sockets

Messaging: Real-time chat with file **uploads** and **instant** document scanning.



Mobile Application — Key Features

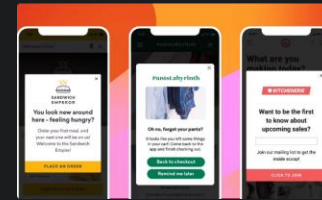


Attendance & Absences

Interactive calendar for daily pointage logging with offline caching and automatic sync on reconnection

Real-Time Chat & Documents

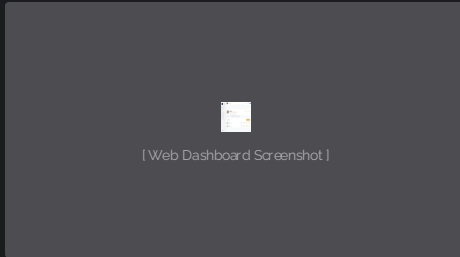
Instant messaging with file attachments, shared document panels, and an integrated camera-based document scanner



Disciplinary Warnings

Categorised letter templates with Gmail integration for formal dispatch to selected employee recipients

Web Administration Platform

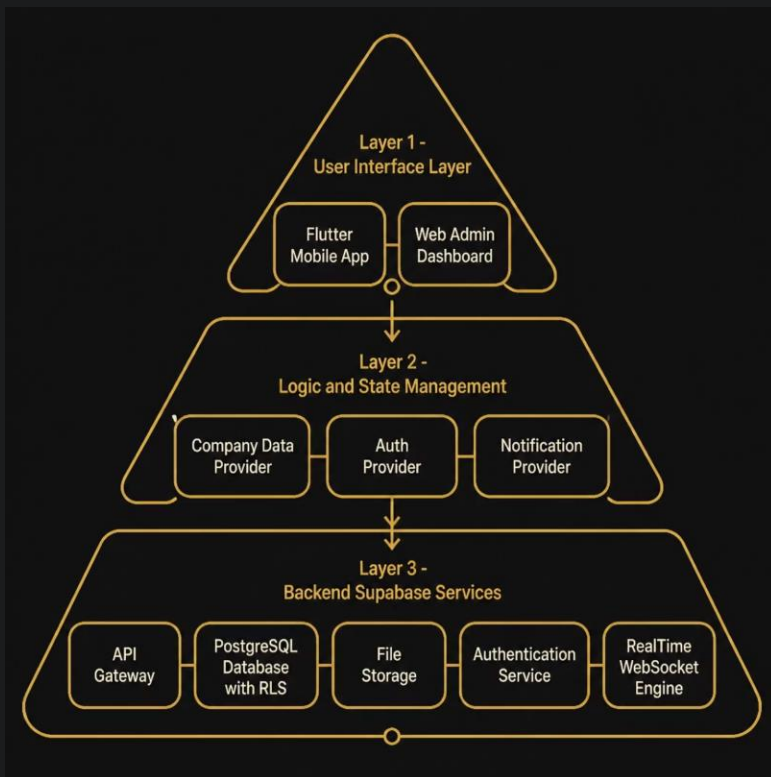


Administrator Workspace

The web platform provides HMI Stars staff with comprehensive oversight across all client companies:

- **Company Directory** — registration, categorisation, and detailed workspace per client
- **Employee Registry** — active staff management with bilingual attendance export (Excel/CSV)
- **Central Messaging** — real-time chat with all clients and shared file archives
- **Global Notes & Reminders** — cross-company task monitoring and urgent alert validation
- **Accounts Management** — role control, manual email verification, and impersonation mode

System Architecture



Client-Server with BaaS

Both Flutter frontends communicate with a centralised Supabase backend via **REST API** and **WebSocket subscriptions**. PostgreSQL Row Level Security enforces per-company data isolation at the database level — no company can access another's records.

Key security guarantee: JWT tokens carry the company ID; RLS policies restrict all queries to matching records only.

Core Technology Stack

Framework selection for cross-platform and data reliability:



Flutter

Single source codebase for Android, iOS, and Web platforms.



Supabase

Managed cloud BaaS with WebSockets streams, Storage, and Auth.



PostgreSQL

Relational storage database protecting system entries.

Password Recovery Sequence

Secure authentication routing via URL hash parameters:



Reset Request

User requests recovery email link in mobile app, generating a secure token.



Redirect Token

Link redirects user back to app with an access token hash parameter.



Session Verify

Flutter captures parameters and validates session with BaaS database.



Tuesday 9 June

Hello, Abdelli Hassen!**2****Staff**

Active employees

**1****Unpointed**

To point today

REMINDERS**Daily Attendance**

Perform the attendance check now! 1 employee has not been pointed yet.

1**Undefined Absences**

Classify the absences now! 1 absence of type "Other" must be specified as soon as possible.

1**Warnings**

Don't forget to send warnings! 15 templates are ready to be sent.

15

Attendance Sync Flow

Offline-tolerant clock-ins with automatic synchronization:

Local Write

Log is written to local SQLite database when offline.

Cloud Sync

Batch sync uploads SQLite queue once internet returns.

Engine Validate

Server verifies timestamp **consistency before** writing to ledger.

Absence Filing Sequence

Transactional consistency from mobile request to database logs:



Submit Request

Manager logs absence leave on mobile with description.



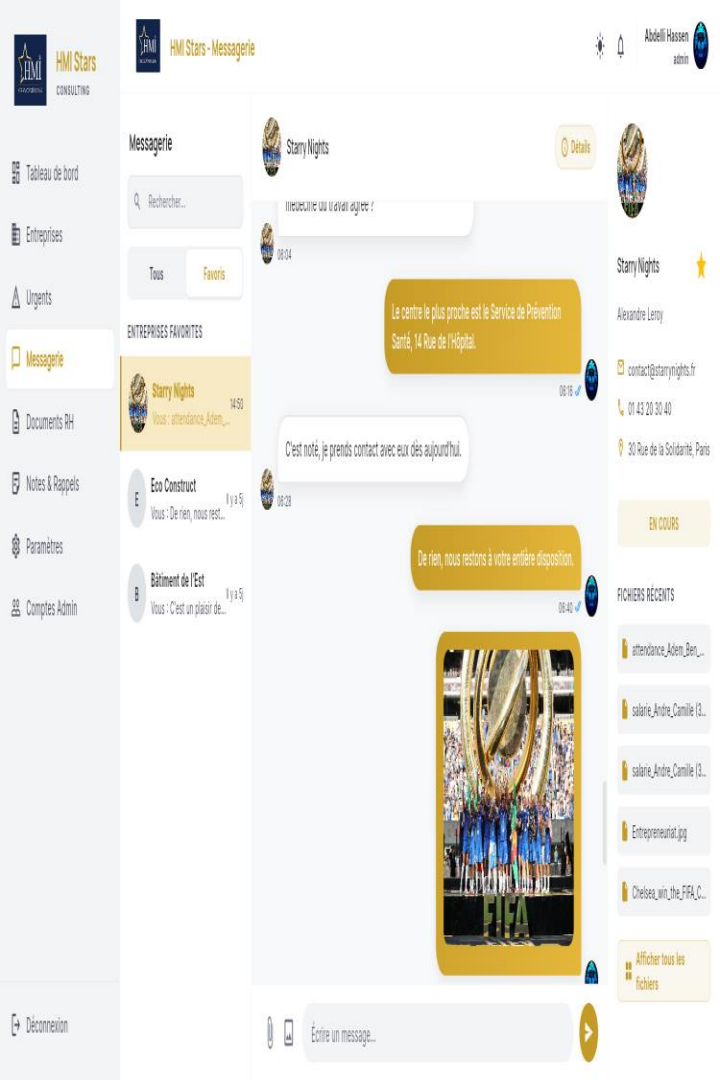
RLS Validation

Database RLS validates user company ID.



Instant Broadcast

Record is inserted and broadcast to Web Admin via sockets.



Real-time Messaging Flow

WebSocket broadcasts using PostgreSQL Write-Ahead Logs:

Zero Polling

Eliminates HTTP polling completely, saving battery and server resources.

Push Replication

WAL logs push updates immediately through active WebSocket connections.

Instant Chat

Messages render in less than **200ms in admin** and mobile panels.

Structural Design: Classic Classes

Conceptual class models, attributes, and object-oriented methods:



Company Model

- companyName: String
- address: String
- getEmployees(): List



Employee Model

- name: String
- contractType: Enum
- getLeaves(): List



User Model

- role: UserRole
- preferences: Map
- authenticate(): boolean

UUID Primary Keys

Status Enums

Ensures type check compliance
at database level.

Tenant Anchors

company_id foreign key **references** on all tables.

Database Design (Physical)

PostgreSQL tables constructed in the cloud instance:



companies Table

• id: uuid PRIMARY KEY • name: text •
address: text



employees Table

• id: uuid PRIMARY KEY • company_id:
uuid FK • full_name: text



attendance Table

• id: uuid PRIMARY KEY • employee_id:
uuid FK • check_in: timestampz



Gestion des Comptes



Tableau de bord

Entreprises

Urgents

Messagerie

Documents RH

Notes & Rappels

Paramètres

Comptes Admin

Comptes Utilisateurs

Gérez les rôles et les accès de tous les utilisateurs de la plateforme.

Rechercher par nom ou email...

 Abdell Hassen ibdelhassen7@gmail.com	Créé le 12/5/2026		ADMIN	 
 tgef ibdelhassen87@gmail.com	Créé le 6/6/2026		SECRÉTAIRE	 
 Mohamed Sghaier sg5348047@gmail.com	Créé le 9/6/2026		SECRÉTAIRE	 
 osghdyg dimassimal4337@gmail.com	Créé le 13/6/2026		SECRÉTAIRE	 
 jihed bormi abdelhassen7@gmail.com	Créé le 14/6/2026		SECRÉTAIRE	  
 bbbbbbbbbbbbbbbb	Créé le 14/6/2026		SECRÉTAIRE	 

Déconnexion

Database Security

Isolating tenant records via Row-Level Security policies:

Row-Level Security

RLS blocks direct client queries from reading another tenant's files.

Tenant Isolation

Enforced via `auth.jwt()` claims to verify company ID matches query.

Granular Access

Restricts read/write access **based on active** role credentials.

Technical Elaborations: DB Views

Computed joins and virtual tables mapping relational structures:



Employee Analytics

Aggregates worked hours, overtime logs, and leave status counts.



Real-time Chat Feeds

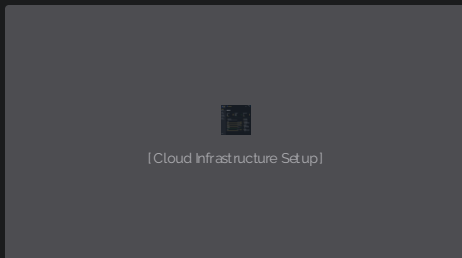
Computed join combining user messages with active company IDs.



Audit Compliance Log

Virtual table tracking validation actions of legal document files.

System Deployment

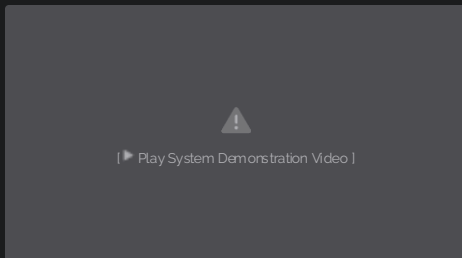


Deployment Pipelines

Distribution pipelines for mobile and web clients:

- **Web Panel** — Hosted on Vercel utilizing global Edge networks for sub-second page loads.
- **Database Vault** — Hosted on Supabase cloud servers, using WAL logs replication.
- **Mobile App** — Compiled into native binaries (APK/IPA) using Flutter's native compiler.
- **Global Notes & Reminders**

System Demonstration

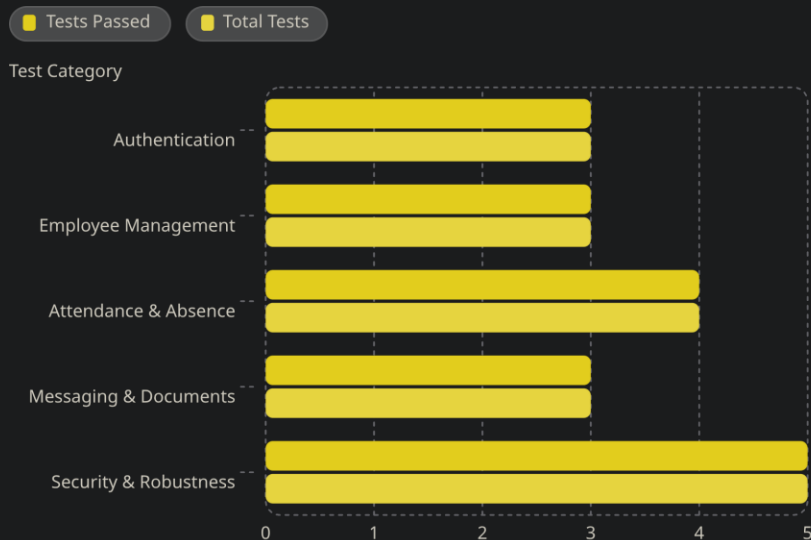


Ecosystem Walkthrough

Watch real-time synchronization between mobile and web clients:

- **1. Offline Pointage** — Field staff clock in offline; timesheets cache locally in SQLite database.
- **2. Background Sync** — Automatically uploads offline queue to PostgreSQL once connection restores.
- **3. Real-time Sockets** — Chat messages and file uploads broadcast instantly via WebSockets.
- **Global Notes & Reminders**

Testing & Validation Results



18/18 Test Cases Passed

All functional tests passed across both platforms, covering authentication, employee CRUD, attendance logging, real-time messaging, document scanning, RLS isolation, and offline caching.

Usability Evaluation

Ten participants rated all five criteria — **Learnability**, **Responsiveness**, **Feedback**, **Navigation Consistency**, and **Real-time Reliability** — as **Good**.

Conclusion & Future Work

The HMI Stars ecosystem successfully centralises communication, document traceability, and employee management — replacing fragmented manual processes with a secure, real-time platform.

Short Term

Automated email digests and daily status reports via Supabase Edge Functions

Medium Term

Automated PDF payslip generation from approved monthly attendance logs

Long Term

AI-powered document classification and OCR to streamline document intake

FLUTTER

SUPABASE

POSTGRESQL

RLS

REAL-TIME

HMI Stars: Web & Mobile Application

Thank you for your attention.

DEVELOPED BY HASSEN ABDELLI

Hassen Abdelli & Yassine Elhadjand recommendations.